ModelIt!

Student Activity Pack

Who Eats Who? Energy Flow Through Food Webs

Tracing Energy from Sunlight to Top Predators

Grade 06L2

NGSS MS-LS2-3, MS-LS2-4

Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think happens to energy when a rabbit eats grass?

2 Why do you think there are more mice than hawks in most ecosystems?

3 Draw a simple food chain with at least 3 organisms. Where does the energy come from?

4 If all the wolves disappeared from Yellowstone, what do you think would happen?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Solar Energy Input | Producer Biomass | Primary Consumer Biomass | Secondary Consumer Biomass | Available Energy at Each Level | Decomposer Return Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Balanced Ecosystem

Moderate solar energy, all trophic levels present

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Solar Shutdown

Lock Solar Energy to 10%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Predator Removal

Remove secondary consumers

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tertiary Consumer Biomass — Top predators like eagles or sharks that eat secondary consumers, receiving only 0.1% of original solar energy

Nutrient Concentration — The amount of recycled minerals and nitrogen available in the soil for plant uptake

Competition Intensity — How strongly organisms at the same trophic level compete for limited resources

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If the Sun provides enough energy to power all life on Earth, why are there so few top predators?
Use evidence from your simulation to support your answers.

1 Why can't a food web support an infinite number of trophic levels?

2 How is energy flow different from nutrient cycling in an ecosystem?

3 What would happen to an ecosystem if all decomposers suddenly disappeared?

4 Why does removing a top predator sometimes HELP producers more than removing herbivores?

5 How does the 10% rule connect to the shape of an energy pyramid?

8 STEM Challenge: Design a Balanced Ecosystem

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YOUR ENGINEERING MISSION

Design a self-sustaining ecosystem for a sealed terrarium that maintains energy flow and nutrient cycling for at least 6 months.

THE SCENARIO

A space agency is designing sealed life-support ecosystems for a Mars colony. Your ecology team must design a small-scale model ecosystem that demonstrates balanced energy flow and nutrient recycling without any outside inputs except light.

Engineering Design Thinking

1. WHAT SPECIES WILL REPRESENT EACH TROPHIC LEVEL IN YOUR TERRARIUM?

2. HOW WILL YOU ENSURE ENOUGH PRODUCER BIOMASS TO SUPPORT THE CONSUMERS YOU INCLUDE?

3. WHAT DECOMPOSERS WILL YOU ADD AND WHY ARE THEY CRITICAL FOR A SEALED SYSTEM?

4. HOW WILL YOU MONITOR THE HEALTH OF YOUR ECOSYSTEM OVER TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Ecologists and Conservation Biologists study how energy and nutrients flow through ecosystems to protect biodiversity and design wildlife management plans. They earn $55,000-$110,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In a grassland food web, producers capture 10,000 kJ of solar energy. Using the 10% energy transfer rule, how much energy is available to secondary consumers?

- **A.** 5,000 kJ because half the energy transfers to each level
- **B.** 1,000 kJ because only 10% reaches secondary consumers
- **C.** 100 kJ because only 10% transfers at each step, and secondary consumers are two steps above producers
- **D.** 10 kJ because energy is divided equally among all four trophic levels

2 In the food web model, what happens when secondary consumers are removed from the ecosystem?

- **A.** Primary consumer biomass decreases because they lose their food source
- **B.** Primary consumer biomass increases because nothing is eating them, which then causes producer biomass to crash from overgrazing
- **C.** Producer biomass increases because more sunlight reaches the ground
- **D.** Nothing changes because secondary consumers do not affect lower trophic levels

3 Which statement best explains why energy flows through an ecosystem but does not cycle?

- **A.** Predators at the top of the food web absorb all the energy so none is left to cycle
- **B.** Energy is converted to heat at every trophic level through metabolism, and heat radiates away and cannot be recaptured by organisms
- **C.** Plants can only capture energy once, so it cannot be reused
- **D.** Decomposers destroy the remaining energy when they break down dead organisms

4 A marine biologist notices that removing sea otters from a kelp forest causes the kelp to disappear. Which explanation is supported by the food web model?

- **A.** Sea otters were protecting the kelp from storms
- **B.** Without otters eating sea urchins, urchin populations exploded and devoured the kelp
- **C.** Sea otters were fertilizing the kelp with their waste
- **D.** The kelp died because it needed the shade from sea otter fur

5 Why are decomposers essential to ecosystem survival even though they are not part of the energy pyramid?

- **A.** Decomposers create new energy from dead organisms
- **B.** Decomposers recycle nutrients locked in dead matter back into the soil so producers can use them to grow
- **C.** Decomposers eat predators and keep the food web balanced
- **D.** Decomposers convert heat energy back into chemical energy for producers